



ENG. AYADO

DOMESTIC WATER SUPPLY SYSTEM I

ISCED UNIT CODE: 0732 251 01A

Duration: 100 hours

Relationship to Occupational Standards

This unit addresses the Unit of Competency: **Install Domestic Water Supply I.**

UNIT DESCRIPTION

This unit specifies the competencies required to install water supply system. It involves preparing pipe installation materials, performing domestic pipework and maintaining domestic pipework. It applies in the construction industry.

Summary of Learning Outcomes

S/No	Learning Outcomes	Duration (Hours)
1.	Prepare and quantify pipe installation materials	10
2.	Sketch simple plumbing drawing and symbols	25
3.	Perform domestic pipe work	50
4.	Maintain domestic pipe work	15

Learning Outcomes, Content and Suggested Assessment Methods

Learning Outcome	Content	Suggested Assessment Methods
1. Prepare and quantify pipe installation materials	1.1. Safety Measures 1.1.1. Safe handling of plumbing tools and equipment 1.1.2. Personal protective equipment 1.1.3. Workshop SOP's, rules and regulations. 1.2. Pipe installation materials 1.2.1. Pipes	• Practical • Projects • Observation • Portfolio of evidence • Written assessment • Oral assessment

	1.2.1.1. PPR-Polypropylene random pipes 1.2.1.2. HDPE-High density polyethylene pipes 1.2.1.3. Galvanized iron (G.I) pipes 1.2.1.4. Chlorinated polyvinyl chloride (CPVC) 1.2.1.5. Unplasticized polyvinyl chloride (UPVC) 1.2.2. Caulking supplies 1.2.3. Various types of pipe support 1.2.4. Sandpapers 1.2.5. Threading oil 1.2.6. Thread tape 1.3. Preparation of pipes 1.3.1. Pipe bending 1.3.2. Cold bending 1.3.3. Heat bending 1.3.4. Pipe cutting 1.3.5. Pipe jointing 1.3.6. Pipe threading 1.3.7. Pipe welding 1.4. Quantify Materials and supplies 1.4.1. Measurements 1.4.2. Fittings 1.4.3. Supplies 1.4.4. Material Schedule 1.4.5. Estimation and costing	
2. Sketch simple plumbing drawing and symbols	2.1 Working drawings 2.1.1 Pictorial 2.1.2 Line drawing 2.1.3 Freehand sketching Scale drawings 2.2 Interpretation of Working pipe work drawings 2.3 Measurements and Symbols 2.3.1 Isometric pipe work drawings 2.3.2 Interpret working drawings.	• Practical • Projects • Observation • Portfolio of evidence • Written assessment • Oral assessment

3. Perform domestic pipework	3.1 Assemble Piping tools and equipment 3.1.1 Pipe wrench 3.1.2 Pipe cutter 3.1.3 Hacksaw 3.1.4 Pipe threading machine 3.1.5 diestock 3.1.6 Pipe vice 3.1.7 Files 3.1.8 Screwdrivers 3.1.9 Drill with various sizes of bits 3.1.10 Mallet 3.1.11 Ballpein hammer 3.1.12 Cold chisel 3.1.13 PPR welding machine / Heat Fusion 3.1.14 Pipe bender 3.1.15 Pipes layout. 3.1.16 Terms and concepts 3.1.17 Setting out of the pipes 3.2 Setting out of pipe lay out based on working drawings 3.3 Pipes mounted based on drawing specifications 3.4 Installation of Storage and auxiliary fittings. 3.4.1 Tanks 3.4.2 Cisterns, 3.4.3 Hot water storage 3.4.4 Tee 3.4.5 Unions 3.4.6 Elbows 3.4.7 Adapters 3.4.8 Nipples 3.4.9 Valves 3.4.10 Tank connectors 3.5 Pipe work functionality tests 3.5.1 Water test 3.5.2 Air test 3.5.3 Pressure test 3.6 Housekeeping	• Practical • Projects • Observation • Portfolio of evidence • Written assessment • Oral assessment
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4 Maintain domestic pipe work	4.1 Pipe work repair materials. 4.1.1 Epoxy Putty 4.1.2 Pipe Repair Tape (e.g., silicone tape) 4.1.3 Repair Clamps (metal or stainless steel) 4.1.4 Slip Couplings or Repair Couplings 4.1.5 PVC or CPVC Cement and Primer 4.1.6 Pipe Patch Kits 4.1.7 Replacement Fittings and Sealants 4.2 SOP's for pipe work maintenance. 4.3 Tools and equipment assembling 4.3.1 Pipe Cutter 4.3.2 Adjustable Wrench 4.3.3 Pipe Wrench 4.3.4 Tubing Cutter 4.3.5 Hacksaw 4.3.6 Pipe Reamer/Deburring Tool 4.3.7 Plumber's Tape (Thread Seal Tape) 4.3.8 Propane Torch 4.3.9 Screwdrivers 4.3.10 Pliers 4.3.11 Measuring Tape 4.3.12 Plumber's Snake or Auger 4.3.13 Safety Gear (gloves, goggles, masks) 4.4 Pipe work faults repair 4.5 Housekeeping	• Practical • Projects • Observation • Portfolio of evidence • Written assessment • Oral assessment
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Suggested Methods of Delivery

- Practical
- Direct Instruction
- Discussion
- Demonstration
- Trade projects
- Site visits

Recommended Resources for 25 Trainees

s/no.	Category	Resource Description/ Specifications	Quantity	Recommended Ratio(Item: Trainee)
1.1	Training Materials	Plumbing Principles and Practices Textbooks/online	25	1:1
1.2		Domestic Water Supply Installation Guides/online	25	1:1
1.3		Technical Manuals (manufacturer's guides)	25	1:1
1.4		Plumbing Level 3 Workbooks	25	1:1
1.5		Sample Blueprints for Water Supply Layout	25	1:1
1.6	Learning facilities and Infrastructure	Theory room (10m*8m)	1	1:25
1.7		Workshop (18m*12m)	1	1:25
1.8	Tools	Pipe Cutter and Tubing Cutter	25	1:1
1.9		Adjustable Wrench and Pipe Wrench	25	1:1
1.10		Deburring Tool	25	1:1
1.11		Screwdrivers and Pliers Set	25	1:1
1.12		Measuring Tape	25	1:1
1.13		Pipe Reamer	25	1:1
1.14		Hacksaw	25	1:1
1.15		Mallet	25	1:1
1.16		Ball pein hammer	25	1:1
1.17		Cold chisel	25	1:1
1.18		Diestock	5	1:5
1.19		Pipe vice	5	1:5
1.20	Equipment	Pipe Bending Machine	5	1:5
1.21		Pressure Testing Kit	5	1:5
1.22		Water Pump Model	1	1:25 (demonstration only)
1.23		Pipe threading machine	5	1:5
1.24		Drill with various sizes of bits	5	1:5
1.25		PPR welding machine / Heat Fusion/ Propane Torch (for supervised use)	5	1:5
1.26	Materials	PVC, PEX, Copper, Galvanized Steel Pipes	25 pieces	1:1
1.27		Fittings (Elbows, Tees, Reducers, Couplings, Tank connectors)	25 pieces	1:1
1.28		Valves (Gate, Ball, Check)	25 pieces	1:1

1.29		Thread Seal Tape and Pipe Joint Compound	25 pieces	1:1
1.30		Pipe Insulation Material	25 pieces	1:1
1.31	PPE's	Gloves, Goggles, Ear Protection, Masks	25 pieces	1:1